PROBLEM3:

PROBLEM STATEMENT:

Hospital Billing and Patient Classification: Process consultation, diagnostic, and treatment charges, apply conditional concessions, validate billing constraints, and classify patients according to a supplied payment policy

OBJECTIVE:

To develop a Python program that calculates hospital bills by adding consultation, diagnostic, and treatment charges. The system applies conditional concessions, validates billing constraints, calculates the final payable amount, and classifies patients according to a payment policy.

PROGRAM:

```
# Hospital Billing and Patient Classificationname = input("Enter patient name: ")patient_type = input("Enter patient type (General/Senior): ")consultation = float(input("Enter consultation charges: "))diagnostic = float(input("Enter diagnostic charges: "))treatment = float(input("Enter treatment charges: "))# Validate billing constraintsif consultation < 0 or diagnostic < 0 or treatment < 0:  print("Invalid billing: Charges cannot be negative.")else:  total = consultation + diagnostic + treatment # Apply conditional concession  if patient_type.lower() == "senior":  concession = total * 0.20  elif total >= 10000:  concession = total * 0.10  else:  concession = 0  final_bill = total - concession  # Classify according to payment policy if final_bill <= 5000:  category = "Low Payment"  elif final_bill <= 15000:  category = "Medium Payment"  else:  category = "High Payment"  print("\n--- Hospital Bill ---")  print("Patient Name:", name) print("Total Charges: Rs.", round(total, 2))  print("Concession: Rs.", round(concession, 2))  print("Final Bill: Rs.", round(final_bill, 2))  print("Patient Classification:", category)
```

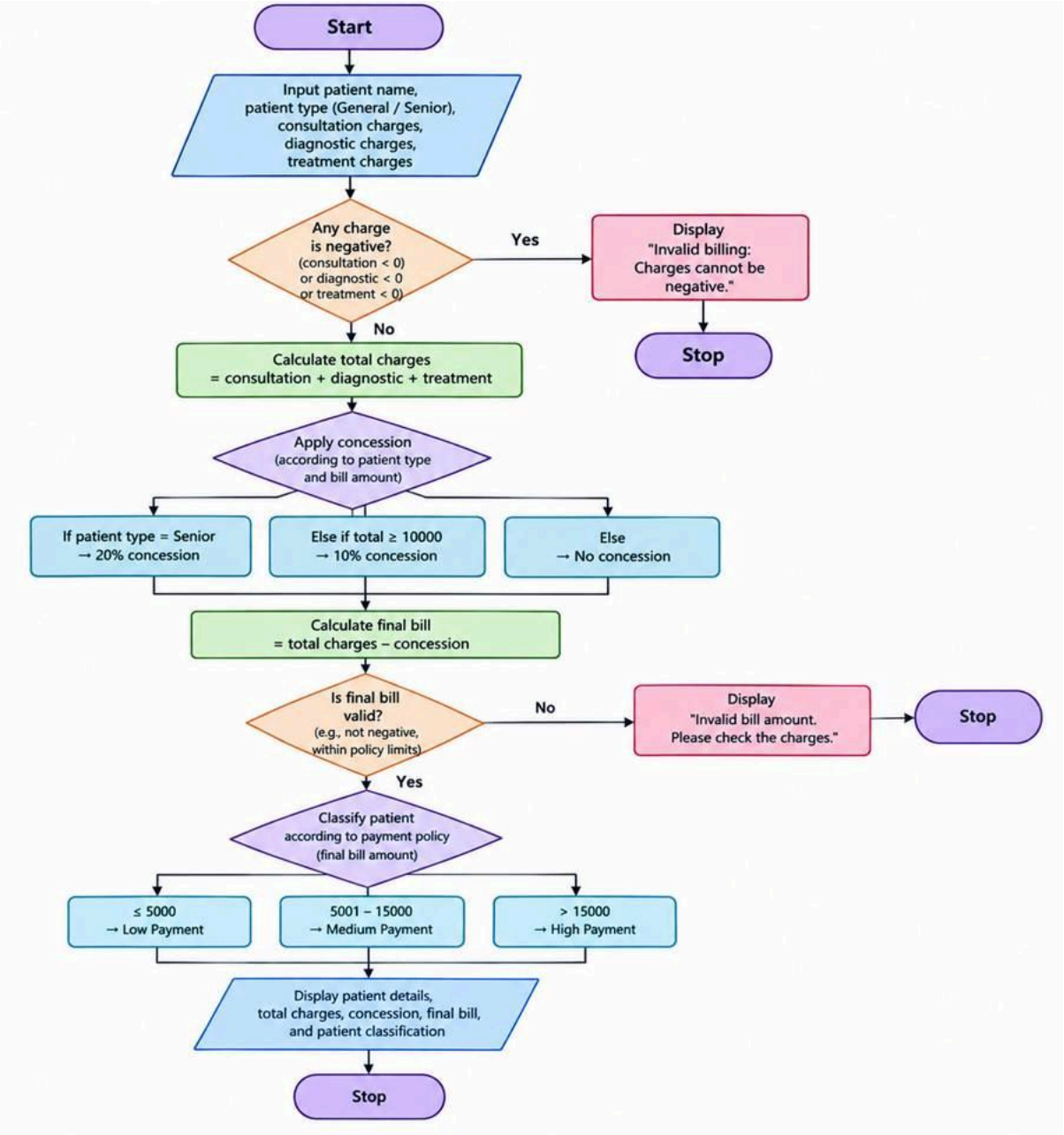
ALGORITHM:

1. Start.
2. Input the patient's name and patient type.
3. Input consultation, diagnostic, and treatment charges.
4. Check whether any charge is negative.
5. If any charge is negative, display an invalid billing message.
6. Otherwise, calculate total charges by adding all three charges.
7. Apply a concession:
    - Senior patient: 20% concession.
    - Other patient with a bill of ₹10,000 or more: 10% concession.

- Otherwise: no concession.

8. Calculate the final bill by subtracting the concession from total charges.
9. Classify the patient according to the final bill:
   - ₹5,000 or less: Low Payment.
   - ₹5,001–₹15,000: Medium Payment.
   - Above ₹15,000: High Payment.
10. Display the bill details and patient classification.
11. Stop.

FLOWCHART:



PSEUDOCODE:

```
BEGIN
        INPUT patient_name, patient_type
        INPUT consultation, diagnostic, treatment

        IF any charge < 0 THEN
        DISPLAY "Invalid Billing"
        ELSE
        total = consultation + diagnostic + treatment

        IF patient_type = "Senior" THEN
        concession = total * 20 / 100
        ELSE IF total >= 10000 THEN
        concession = total * 10 / 100
        ELSE
        concession = 0
        END IF

        final_bill = total - concession

        IF final_bill <= 5000 THEN
        category = "Low Payment"
        ELSE IF final_bill <= 15000 THEN
        category = "Medium Payment"
        ELSE
        category = "High Payment"
        END IF

        DISPLAY total, concession, final_bill
        DISPLAY category
```

```
        END IF
END
```

OUTPUT:

Enter patient name: Ravi

Enter patient type (General/Senior): General

Enter consultation charges: 1000

Enter diagnostic charges: 4000

Enter treatment charges: 6000

--- Hospital Bill ---

Patient Name: Ravi

Total Charges: Rs. 11000.0

Concession: Rs. 1100.0

Final Bill: Rs. 9900.0

Patient Classification: Medium Payment